

Super-Eddington Little Blue Dots Over-Reionize Helium

JWST's faint AGN may complete helium reionization too early unless only a small fraction are super-Eddington.

arXiv:2608.28745 — Super-Eddington Little Blue Dots May Reionize Helium Too Early

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Background

JWST found many faint broad-line AGN at $z \gtrsim 4$, dubbed Little Blue Dots. Their blue continua and compact sizes suggest super-Eddington accretion with strong extreme-UV but weak X-rays.

Do these super-Eddington LBDs reionize helium too early, conflicting with observations?

What this paper is selling

The paper tests whether super-Eddington LBDs can dominate helium reionization without violating timing constraints. It finds that only $\lesssim 10\%$ can be super-Eddington, unless escape fractions are tiny.

Key results to remember

- He II reionization ending at $z \approx 3$ disfavors mostly super-Eddington LBDs.
- Super-Eddington fraction must be $\lesssim 10\%$ for fiducial escape fraction 15%.
- Majority super-Eddington requires escape fractions $\lesssim 1.5\%$, contradicting observations.

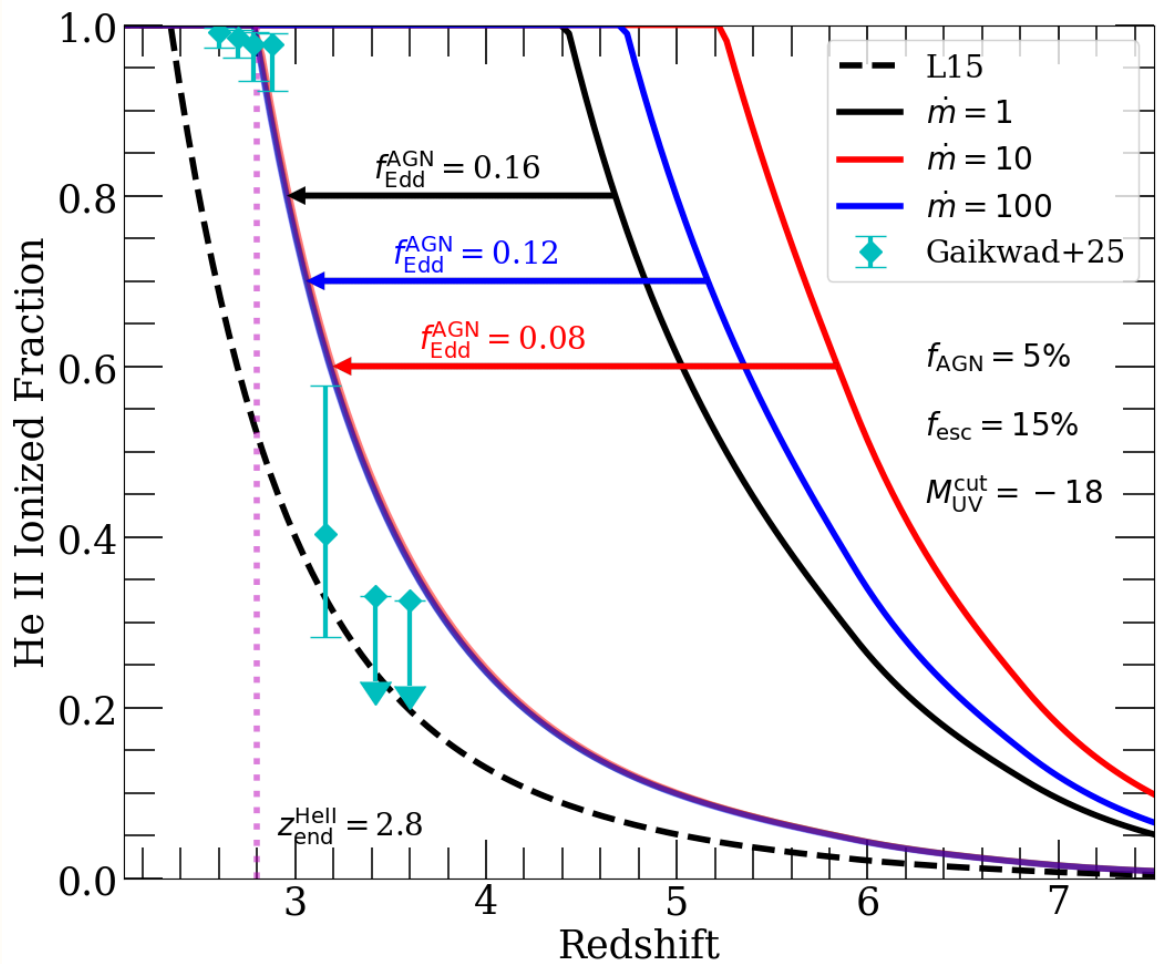


Figure 3 | He II reionization histories show that super-Eddington models complete reionization too early, conflicting with $z \approx 2.8$ constraints.

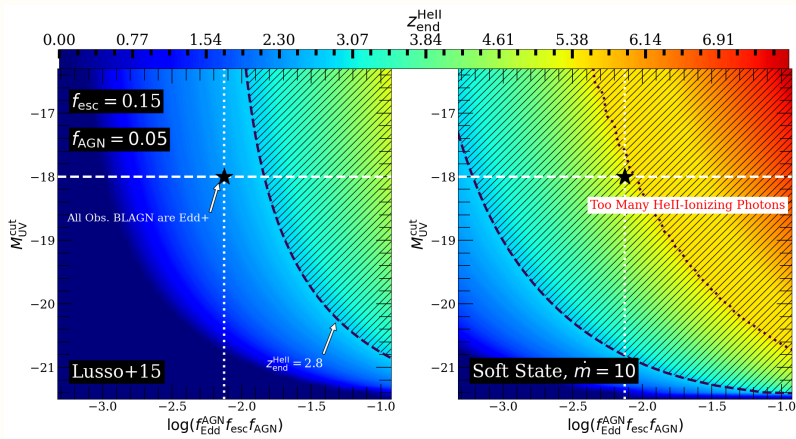


Figure 4 | Excluded parameter space reveals that high super-Eddington fractions and escape fractions are ruled out by helium reionization timing.